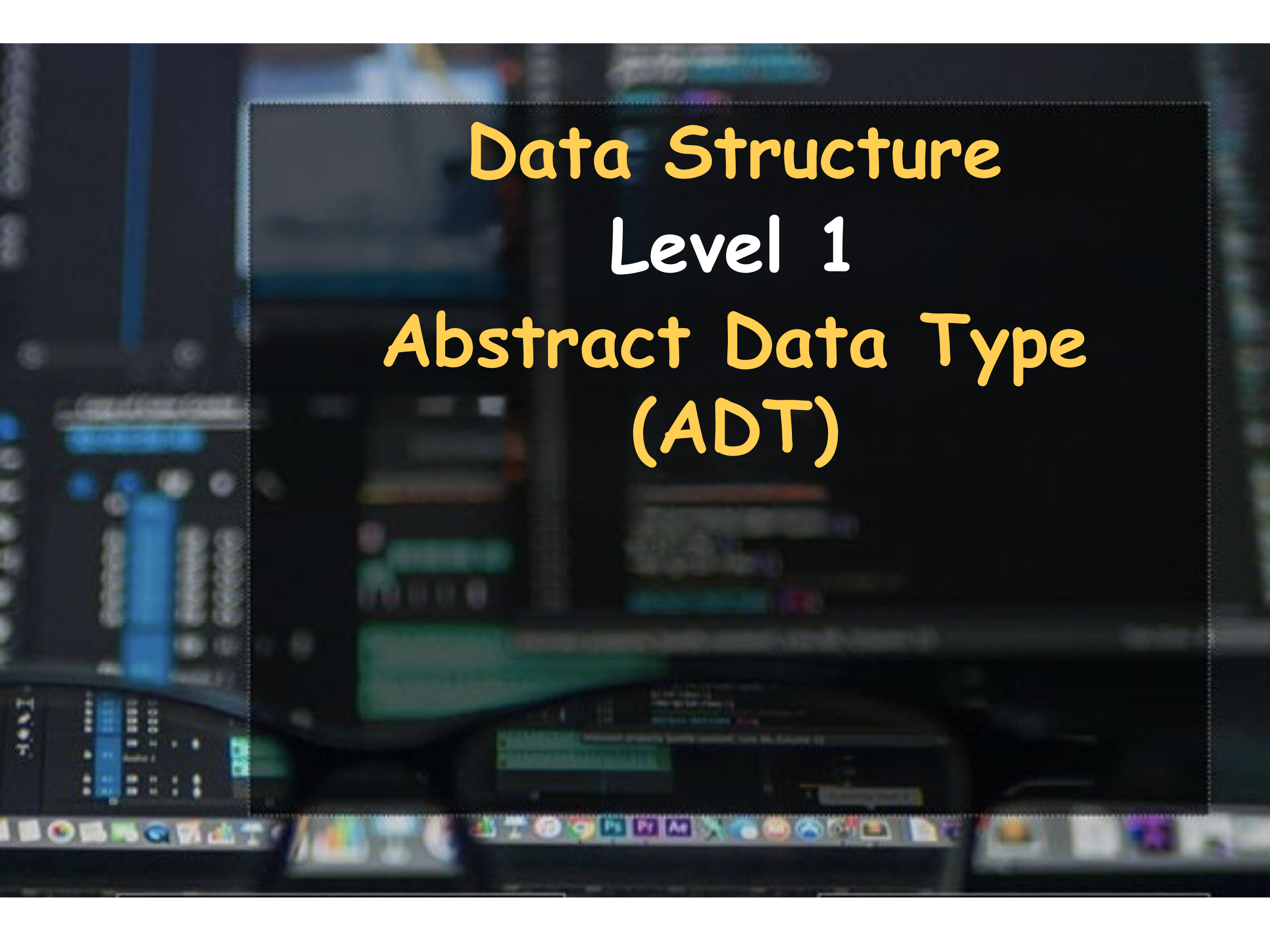
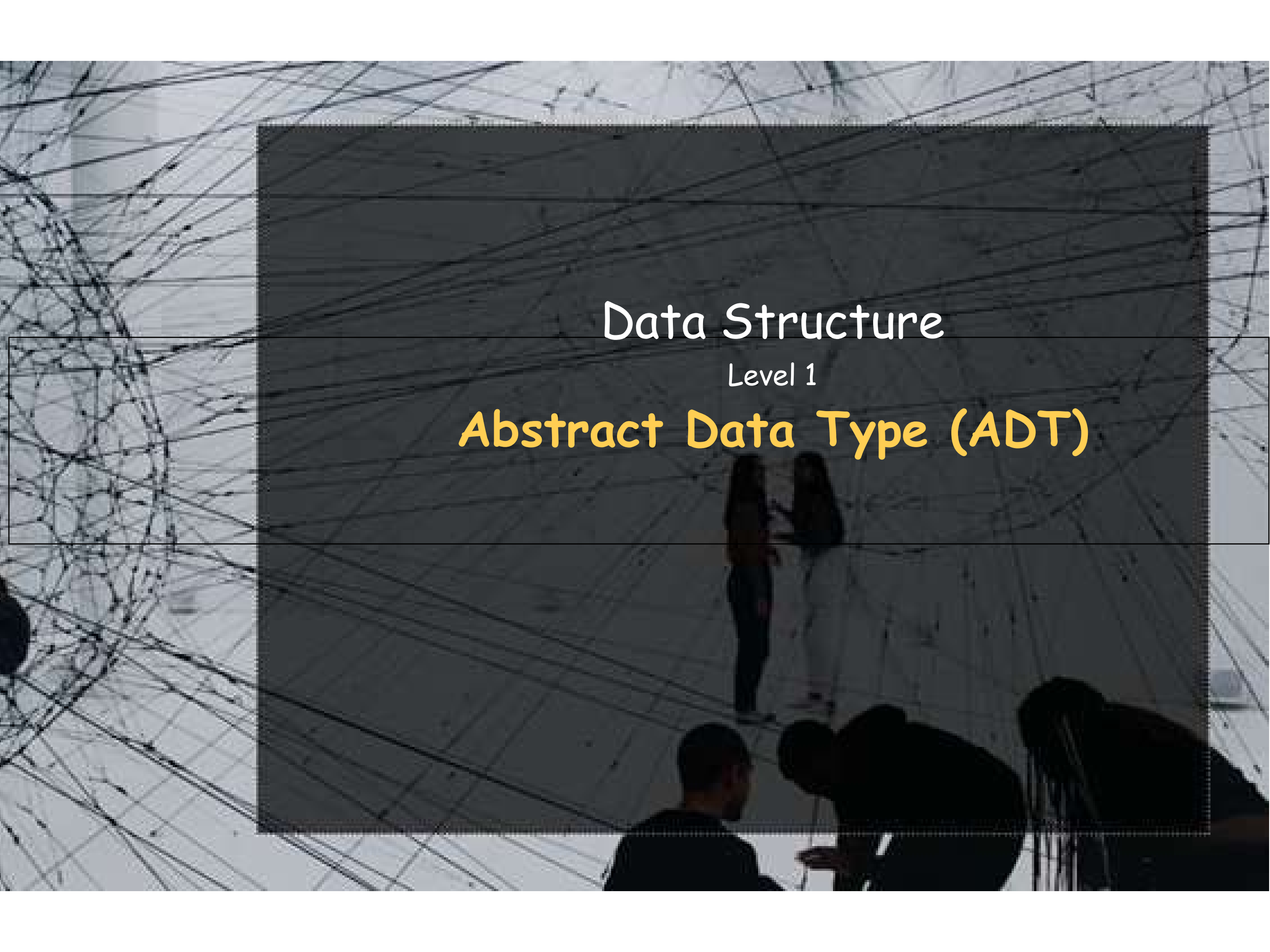




Data Structure

Level 1

Abstract Data Type (ADT)



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Level 1

Abstract Data Type (ADT)

Definition

⌞ An Abstract Data Type (ADT) is a logical description of a data type that defines what operations can be performed on the data, without specifying how those operations are implemented.

Explanation

◆ An ADT focuses on what the data structure does, not how it works internally.

▣ It defines:

▨ The type of data

▨ The operations that can be performed

▨ The behavior of those operations

▣ It does not define:

▨ How the data is stored

▨ The exact code used to implement the operations

▨ Which specific data structure must be used

◆ Delete the last node using delete.

Procedures (ADT)

1. Define the Data

-  Decide what type of data the ADT represents.

💡 Example:

Stack → stores elements

2. Define the Operations

-  Specify what operations are available.

-  `Push()` → adds an element

-  `Pop()` → removes the top element

-  `Top()` → returns the top element

-  `IsEmpty()` → checks whether the stack is empty

Procedures (ADT)

Choose an Implementation

The ADT can be implemented using different data structures.

Array

Linked List

Result: The user can use the same operations without needing to know the internal implementation.

Example

Stack ADT

Stack

Push()

Pop()

Top()

IsEmpty()

It can be implemented using an Array:

Array → Stack ADT

Or using a Linked List:

Linked List → Stack ADT

The interface and behavior remain the same even though the implementation is different.

Note

◆ ADT = What

◆ Data Structure = How

💡 Example:
Stack ADT



What can we do?



Push, Pop, Top

Implementation



How do we do it?



Array or Linked List

💡 Example

Stack ADT



Array

Linked List

➡ Both can provide the same Stack operations.

Comparison

Abstract Data Type (ADT)

- ✓ Defines what operations are available
- ✓ Focuses on behavior
- ✓ Hides implementation details
- ✓ Can have multiple implementations

Data Structure

- ✓ Defines how data is stored
- ✓ Focuses on implementation
- ✓ Determines how operations are performed
- ✓ Examples: Array, Linked List, Tree

⊘ Warning

⚠ Do not confuse an ADT with a data structure

▣ An ADT describes the behavior and operations

▤ A data structure describes how the data is organized and stored

✦ Summary

- ✓ ADT defines what operations can be performed
- ✓ Focuses on behavior, not implementation
- ✓ Hides internal implementation details
- ✓ One ADT can have multiple implementations
- ✓ ADT = What, Data Structure = How

Thank you 😊